

Problem 1: The Geometry of Momentum

In Homework 1, you analyzed the eigenvalues of the momentum update matrix. Now, we will apply that intuition to a specific 1-dimensional "toy" problem.

We are trying to minimize $\mathcal{L}(w) = \frac{1}{2}\sigma^2 w^2$ using Gradient Descent with Momentum. The update rule is:

$$\begin{aligned}w_{t+1} &= w_t - \eta z_{t+1} \\z_{t+1} &= (1 - \beta)z_t + \beta \nabla \mathcal{L}(w_t)\end{aligned}$$

(a) **[System Dynamics]** We can write the state evolution as a linear system:

$$\begin{bmatrix} z_{t+1} \\ w_{t+1} \end{bmatrix} = R \begin{bmatrix} z_t \\ w_t \end{bmatrix}$$

Fill in the four entries of the matrix R in terms of β , η , and σ .

Problem 2: The "Lazy" Regularizer

Your teammate wants to train a linear model $Xw \approx y$ with **Weight Decay**, but the specialized hardware accelerator you are using only supports standard Stochastic Gradient Descent (SGD) on the Mean Squared Error (MSE) loss, with no option to add a separate L_2 penalty term to the loss function.

The available hardware API is: `HardwareSGD(input_batch, target_batch, learning_rate)`

You recall from Homework 1 that you can mimic regularization by injecting noise.

(a) **[Input Noise]** You propose adding random Gaussian noise $\xi \sim \mathcal{N}(0, \nu^2 I)$ to the **input** x during training. If you want to achieve an effective Weight Decay equivalent to minimizing:

$$\mathcal{L}_{reg}(w) = \frac{1}{m} \|Xw - y\|^2 + \lambda \|w\|^2$$

What variance ν^2 should you choose for your noise? Express ν^2 in terms of λ and/or the dataset size m .

(b) **[Label Noise]** Your teammate suggests that instead of adding noise to the input, it might be computationally cheaper to add noise to the **labels** y . They propose: $\tilde{y} = y + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \gamma^2)$.

Does this achieve L_2 regularization (Weight Decay) on w ? If yes, what is the effective λ ? If no, mathematically describe what effect this noise has on the expected loss function.

Problem 3: SGD Memory

We often assume SGD converges to the minimum norm solution in the "interpolation regime" (where X is wide and we can fit data perfectly). However, this depends strictly on initialization.

Consider a simple 2D problem with 1 data point:

$$X = [1, \quad 2] \quad y = [5]$$

We want to solve $Xw = y$, i.e., $w_1 + 2w_2 = 5$.

(a) **[The Standard Case]** If we initialize $w_{init} = [0, \quad 0]^\top$, calculate the final solution w_∞ that SGD will converge to.

(b) **[The "Bad" Initialization]** Suppose we initialize $w_{init} = [5, 0]^\top$ (which is NOT the solution, since $5 + 2(0) = 5$ holds... wait, actually check if this is a solution. $1(5) + 2(0) = 5$. Yes, it is a valid solution!).

Since $w_{init} = [5, 0]^\top$ is already a solution (loss is zero), SGD will not move. Is this the **minimum norm** solution? If not, calculate the squared Euclidean distance between this solution and the true minimum norm solution from part (a).

(c) **[The General Case]** Suppose we initialize at a non-solution point $w_{init} = [4, 0]^\top$. SGD updates always add a vector proportional to the row(s) of X . Using the fact that $w_\infty - w_{init} = \alpha X^\top$ for some scalar α , calculate the final vector w_∞ .

Problem 4: Bayesian Twists

In Homework 1, you proved that Ridge Regression is equivalent to MAP estimation with a Gaussian Prior $w \sim \mathcal{N}(0, \tau^2 I)$ and Gaussian Noise $n \sim \mathcal{N}(0, \sigma^2 I)$.

We are going to modify the probabilistic assumptions. For each modification below, identifying the new corresponding Loss Function terms.

Structure your answer in the form:

$$\operatorname{argmin}_w ||Xw - y||^2 + \dots$$

(Fill in the dots).

(a) **[Biased Prior]** We believe the weights should be close to a specific "guess" vector w_0 , rather than 0.

$$\text{Prior: } w \sim \mathcal{N}(w_0, \tau^2 I)$$

What is the new Regularization term?

(b) **[Heteroscedastic Noise]** We trust some data points more than others. The noise for the i -th data point has its own variance σ_i^2 .

$$y_i = w^\top x_i + n_i, \quad n_i \sim \mathcal{N}(0, \sigma_i^2)$$

How does the **Reconstruction Loss** term ($\|Xw - y\|^2$) change? Write the new sum over $i = 1 \dots m$.